

JONATHAN KUNIARDI

SOFTWARE ENGINEERING INTERN — AI & PYTHON

North Jakarta, Indonesia | kuniardijonathan@gmail.com
linkedin.com/in/jonathankuniardi | fjgb22.github.io | github.com/FJGB22

PROFESSIONAL SUMMARY

Computer Science undergraduate at BINUS University (Intelligent Systems, GPA 3.72/4.00) who builds real-time conversational AI in Python. Architected a voice-driven AI VTuber pipeline (speech recognition, streaming LLM, interruptible text-to-speech) and benchmarked it to locate the true bottleneck, and own the backend of a five-person group ESP32 voice-assistant project. Available for a full-time internship in semesters 6 and 7 through the BINUS Enrichment Program.

EDUCATION

Bachelor of Computer Science | Intelligent Systems | BINUS University, Jakarta Sep 2024 - Present
GPA 3.72 / 4.00 · Semester 5 · Expected graduation 2028

Relevant Coursework: Machine Learning, Natural Language Processing, Artificial Intelligence, Computer Vision, Data Structures, Software Engineering, Embedded Systems; in progress: Speech Recognition, Deep Learning

SMA Permai | Natural Sciences (IPA), Jakarta 2021 - 2024

ORGANIZATIONAL EXPERIENCE

Binusian Gaming (BG) | Regional Secretary, Kemanggisian | Jakarta Jan 2026 - Present

- Administer the formal documentation of a region of ~30 committee members and 100+ members per intake — meeting minutes, event proposals, accountability reports (LPJ), and MoU/MoA.
- Took on the secretary workload from late 2025, months ahead of formal appointment, after more than a year as an activist (since Nov 2024).

BINUS New Student Expo | Booth Lead, Binusian Gaming | Jakarta 2025 - 2026

- Spearheaded the organization's booth two years running — designed the concept, recruited and briefed a crew of under 10, and reached 150+ incoming students at each expo.

BINUS Student Mentoring | Academic Mentor | Jakarta Sep 2025 - Jan 2026

- Guided the 2 consistently active mentees (of 5 assigned) through the full semester while in my own third semester, explaining material on demand across most of their courses instead of a fixed syllabus.

SKILLS

Core: Python (primary); Java, TypeScript, HTML, CSS (foundational); asyncio & threading; LLM integration (Google Gemini API, OpenAI-compatible clients); prompt & persona design; speech-to-text (faster-whisper); text-to-speech (edge-tts)

Tools: Git & GitHub, VS Code, pytest, Ruff, Git pre-commit hooks, Python virtual environments, Figma, Canva, video editing

Methods: Modular architecture, performance benchmarking, test-gated commits, AI-assisted development

Soft Skills: Technical & formal documentation, event leadership, mentoring

Spoken Languages: Indonesian (native), English (intermediate)

PROJECTS

Mika — AI VTuber | Python, asyncio, Gemini API, faster-whisper, edge-tts 2026 - Present
github.com/FJGB22/ai-vtuber · individual, self-initiated

- Architected a real-time voice pipeline of 6 decoupled modules — speech recognition, turn-taking, streaming LLM, interruptible speech output, avatar control, and configuration — so each stage can be developed and replaced independently.
- Engineered sentence-level streaming: replies are split into sentences as tokens arrive and spoken while the model is still generating, rather than waiting for the full response.
- Built a measurement bench (latency, token, jailbreak, and drift tests) and measured a 10.5 s turn, 8.3 s of it audio playback — proving the voice, not the model, was the bottleneck and redirecting the roadmap toward interruptible speech.
- Designed a character persona that withstood jailbreak attempts through in-character consistency rather than refusal rules.

Mika IoT — Voice Assistant Satellite | Python, Google GenAI SDK, ESP32, pytest, Ruff 2026 - Present
Group project (5 members) · Embedded Systems course

- Own the backend: a Python service that streams Gemini replies and keeps multi-turn conversation memory for an Alexa-style ESP32 device, scoped to voice Q&A first so the first version ships.
- Designed a provider-neutral core — history kept in a generic format and translated by one function — so switching LLM vendors means rewriting a single function.
- Enforced code quality with a pre-commit hook that blocks any commit failing lint, format checks, or tests.